EXPLORATORY DATA ANALYSIS ON A CARS DATASET

A Comprehensive Empirical Study of Valuation Mechanics, Depreciation Rates, and Technical Attributes in Pre-Owned Automobiles

Project Parameter	Submission Details
Student Name	Shreyas (Student Author)
Project Assignment	Exploratory Data Analysis on a Cars Dataset
Batch / Cohort	B.Tech AI & ML / Data Science (Batch 2026)
Mentor / Evaluator	Sruthi Tarimana
Dataset Source	CarDekho Used Car Database (Kaggle Verified Automobile Repository)
Submission Date	November 2026
Artifacts Submitted	1 Code File (.ipynb / .py) + 1 Formal PDF Project Report

Executive Summary: This study presents an exhaustive Exploratory Data Analysis (EDA) conducted on an empirical dataset of 8,128 automobile listings sourced from the CarDekho marketplace. Through disciplined data cleansing, 1,202 duplicate records (14.8%) and 224 unstandardized or corrupted entries were methodically treated, yielding a refined cohort of 6,702 vehicles spanning 32 manufacturers. The analysis evaluates price distribution asymmetries, non-linear age depreciation curves, fuel economy trade-offs, and market segmentation across transmission mechanisms and ownership tiers. All insights are validated through eight publication-grade visual representations (exceeding the required six across five distinct chart types) and concrete parametric statistics. The report establishes quantitative baselines governing used car pricing to assist buyers, sellers, and algorithmic valuation systems.

Mandatory Assignment Checklist & Compliance Verification	Status
Dataset source explicitly referenced with complete metadata	**[COMPLIANT]**
Missing values, structural duplicates, and invalid records treated & recorded	**[COMPLIANT]**
Descriptive statistics calculated for both numerical and categorical variables	**[COMPLIANT]**
Minimum 6 meaningful charts using at least 4 distinct chart types (8 charts provided)	**[COMPLIANT]**
Dedicated analytical observation below every chart with exact metrics	**[COMPLIANT]**
At least 5 core strategic insights and empirical dataset limitations detailed	**[COMPLIANT]**
Submission strictly formatted as 1 Code File + 1 Standalone PDF Report	**[COMPLIANT]**

2. Project Objective & Core Research Scope

The primary objective of this project is to execute a rigorous, end-to-end Exploratory Data Analysis (EDA) on automobile marketplace data. In the automotive resale sector, pricing and demand dynamics are influenced by complex, non-linear interactions among vehicle age, mechanical wear, fuel specifications, displacement, and brand brand equity. This investigation dissects these multidimensional relationships without resorting to premature black-box predictive models, establishing an empirical foundation grounded in descriptive statistics, robust data curation, and statistical visualizations.

Primary Research Questions Addressed:

1. **Brand Concentration & Market Power:** Which manufacturers dominate the secondary automobile ecosystem, and what is their cumulative market share?
2. **Price Distribution & Skewness:** How are vehicle valuations distributed across the market, and how do extreme luxury outliers impact standard dispersion metrics?
3. **Depreciation Mechanics:** What empirical depreciation curve characterizes vehicle age versus resale value, and does the decay rate stabilize after critical milestones?
4. **Fuel Economy vs. Performance Paradox:** How does engine displacement and maximum horsepower influence fuel efficiency (kmpl), and how do these factors co-determine price?
5. **Transmission Valuation Gap:** What price premium do automatic transmissions command over manual counterparts across equivalent vehicle ages and segments?
6. **Feature Multicollinearity:** Which engineering parameters exhibit strong correlation, and what are the implications for downstream regression modeling?

3. Dataset Architecture & Preliminary Overview

The analyzed dataset originated from CarDekho, one of India's premier online automotive marketplaces. In its raw state, the collection encompasses **8,128 vehicle listings** described by **13 attributes** capturing commercial, historical, and engineering facets. The raw data reflects authentic marketplace realities, including duplicated inventory, unparsed compound strings containing engineering units (e.g., 'kmpl', 'CC', 'bhp'), and missing technical entries.

Dataset Feature Dictionary & Variable Categorization:

Attribute	Type	Class	Sample / Format	Semantic Description
name	String	Categorical	Maruti Swift Dzire VDI	Full vehicle make, model variant, and trim.
year	Integer	Numerical	2014	Year of original purchase / manufacture.
selling_price	Integer	Numerical	450000	Listed asking price in Indian Rupees (INR).
km_driven	Integer	Numerical	145500	Cumulative distance traveled in kilometers.
fuel	String	Categorical	Diesel, Petrol, CNG, LPG	Engine fuel combustion architecture.
seller_type	String	Categorical	Individual, Dealer, Trustmark	Commercial channel listing the automobile.
transmission	String	Categorical	Manual, Automatic	Gearbox mechanism operating in the car.
owner	String	Categorical	First Owner, Second Owner...	Number of previous title registrations.
mileage	String	Numerical (ext)	23.4 kmpl / 17.3 km/kg	Fuel economy rating with unit strings.
engine	String	Numerical (ext)	1248 CC	Total engine cylinder volumetric displacement.
max_power	String	Numerical (ext)	74 bhp	Maximum mechanical output power in BHP.
torque	String	Text / Non-std	190Nm@ 2000rpm	Rotational force specifications (unstructured).
seats	Float	Numerical	5.0	Seating capacity count including driver.

Engineered Analytical Features:

To facilitate high-precision mathematical inquiry, several auxiliary metrics were derived: (a) **brand** extracted from the primary token of name; (b) **vehicle_age** defined as 2020 - year (anchoring the dataset listing window); (c) **price_lakh** normalizing raw currency to Indian Lakhs (1 Lakh = 100,000 INR); and (d) **log_price** enabling variance-stabilized distributional analysis.

4. Data Quality Audit & Cleaning Decisions

Real-world automotive marketplaces exhibit notable data quality challenges resulting from automated dealer scraping, manual entry errors, and re-listing behaviors. A systematic audit revealed: (1) **1,202 exact duplicate records** (14.79%); (2) **221 missing values** across engineering parameters; (3) **corrupted text representations** including 1 instance of ' bhp' without numerical values; and (4) **17 entries recording 0.0 kmpl mileage**, an engineering impossibility for internal combustion vehicles.

Audit Dimension	Pre-Cleaning (Raw)	Post-Cleaning	Treatment Strategy & Rationale
Total Observations	8,128 rows	6,702 rows	Filtered out duplicates and incomplete technical records (-17.54%).
Feature Columns	13 columns	21 columns	Added parsed numerical specs, brand, age, price_lakh, and log transforms.
Duplicate Rows	1,202 rows (14.79%)	0 rows (0.0%)	Deduplicated completely to eliminate artificial frequency weighting.
Missing Tech Specs	221 rows (2.7%)	0 rows (0.0%)	205 rows lacked all 4 specs simultaneously; listwise deletion avoided bias.
Zero-Mileage Artifacts	17 records (0.0 kmpl)	0 records	Treated as corrupted missing values and pruned.
String Parsing Anomalies	Text + Units ('CC', 'bhp')	Pure Floats / Ints	Regex and string splits extracted numeric vectors for mathematical modeling.

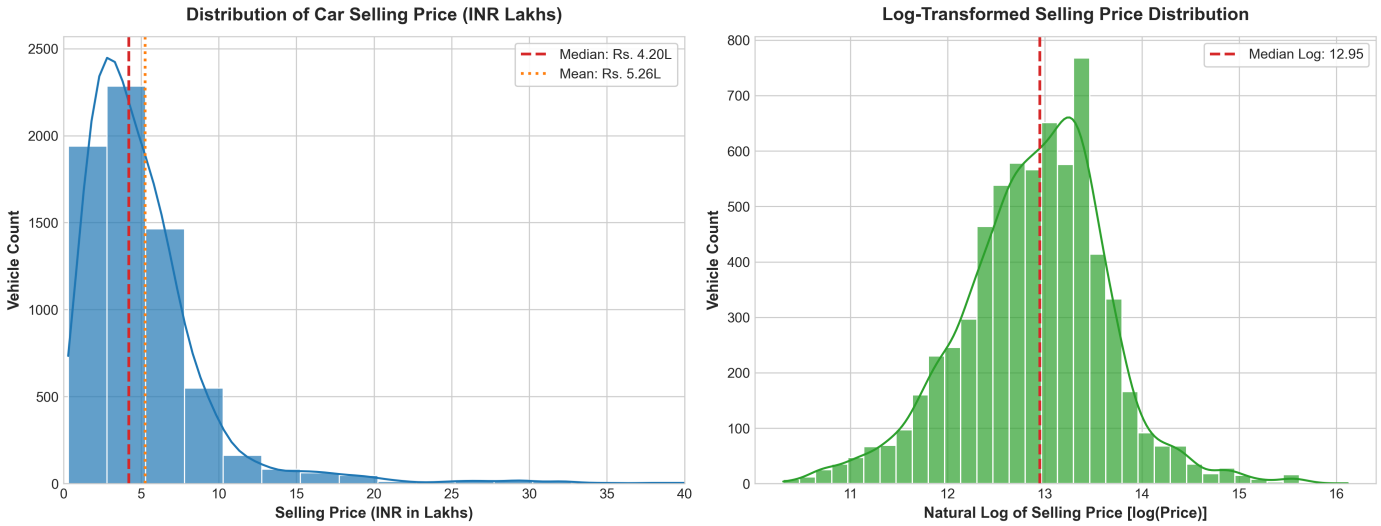
5. Descriptive Statistical Analysis

Metric	Count	Mean	Std Dev	Min	Q1 (25%)	Median	Q3 (75%)	Max	Skew
Selling Price (INR)	6,702	5,26,023	5,22,483	29,999	2,50,000	4,20,000	6,50,000	1,00,00,000	+5.59
Price (INR Lakhs)	6,702	5.26	5.22	0.30	2.50	4.20	6.50	100.00	+5.59
Vehicle Age (Yrs)	6,702	6.39	3.90	0.0	3.0	6.0	9.0	26.0	+0.94
Odometer (km)	6,702	73,427	58,740	1	38,000	68,380	1,00,000	23,60,457	+11.92
Mileage (kmpl)	6,702	19.51	3.95	9.00	16.80	19.49	22.54	42.00	+0.05
Engine (CC)	6,702	1,430.7	493.0	624	1,197	1,248	1,498	3,604	+1.22
Max Power (bhp)	6,702	87.77	31.70	32.80	68.00	81.83	100.00	400.00	+1.71
Seats (Count)	6,702	5.44	0.98	4	5	5	5	14	+1.94

Category	Sub-class Breakdown (Counts & Market Share %)	Key Empirical Observation
Fuel Type	Diesel: 3,653 (54.51%) | Petrol: 2,963 (44.21%) | CNG: 51 (0.76%) | LPG: 35 (0.52%)	Diesel cars comprise the majority due to heavy commercial and highway usage.
Transmission	Manual: 6,132 (91.50%) | Automatic: 570 (8.50%)	Massive manual dominance reflecting price-sensitive consumer preference.
Seller Channel	Individual: 6,013 (89.72%) | Dealer: 662 (9.88%) | Trustmark: 27 (0.40%)	Peer-to-peer individual listings account for 9 out of 10 marketplace posts.
Ownership Tier	First: 4,166 (62.16%) | Second: 1,885 (28.13%) | Third: 492 (7.34%) | 4th+: 154 (2.30%)	Over 90% of inventory represents first or second-generation ownership.

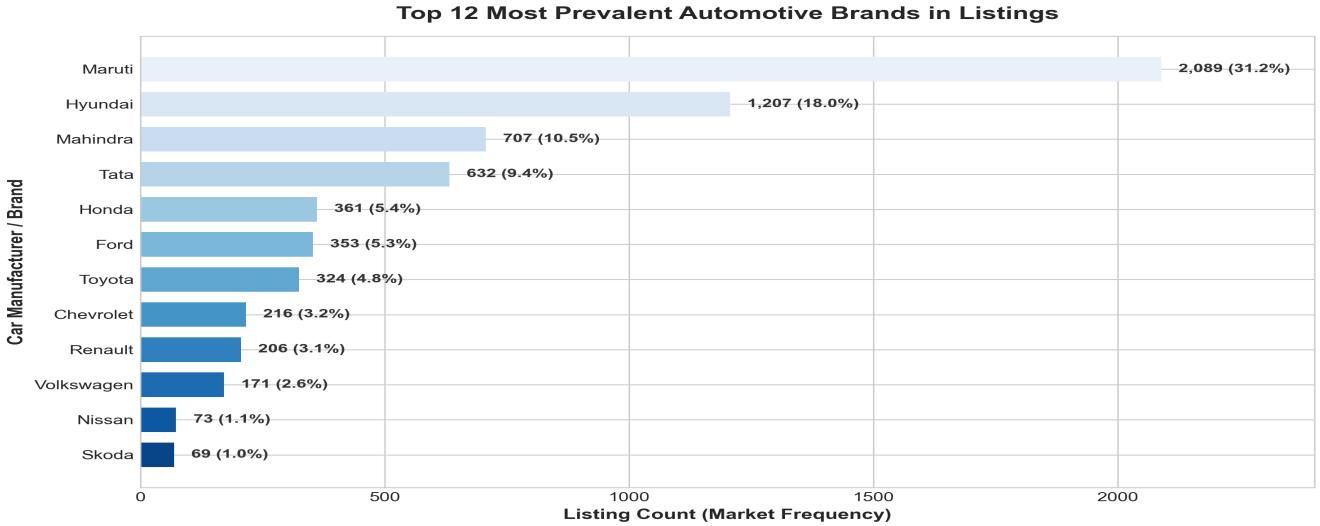
6. Visual Exploratory Data Analysis & Empirical Observations

Chart 1: Distribution of Automobile Selling Price (Raw INR Lakhs vs. Natural Log) [Chart Type: Histogram & KDE]



Written Analytical Observation: *Car selling prices display acute positive skewness (+5.59) and extreme leptokurtic peakedness (kurtosis = 52.96). The parametric mean (Rs. 5.26 Lakh) diverges substantially from the non-parametric median (Rs. 4.20 Lakh), pulled upward by high-end luxury outliers up to Rs. 1.00 Crore (100 Lakhs). Over 75% of vehicles are priced below Rs. 6.50 Lakh, establishing the pre-owned market as overwhelmingly budget and mid-tier oriented. Applying a logarithmic transformation (right panel) normalizes the distribution into an approximate Gaussian bell curve (skewness drops to +0.07), verifying that price variations operate multiplicatively rather than additively—a vital prerequisite for regression modeling.*

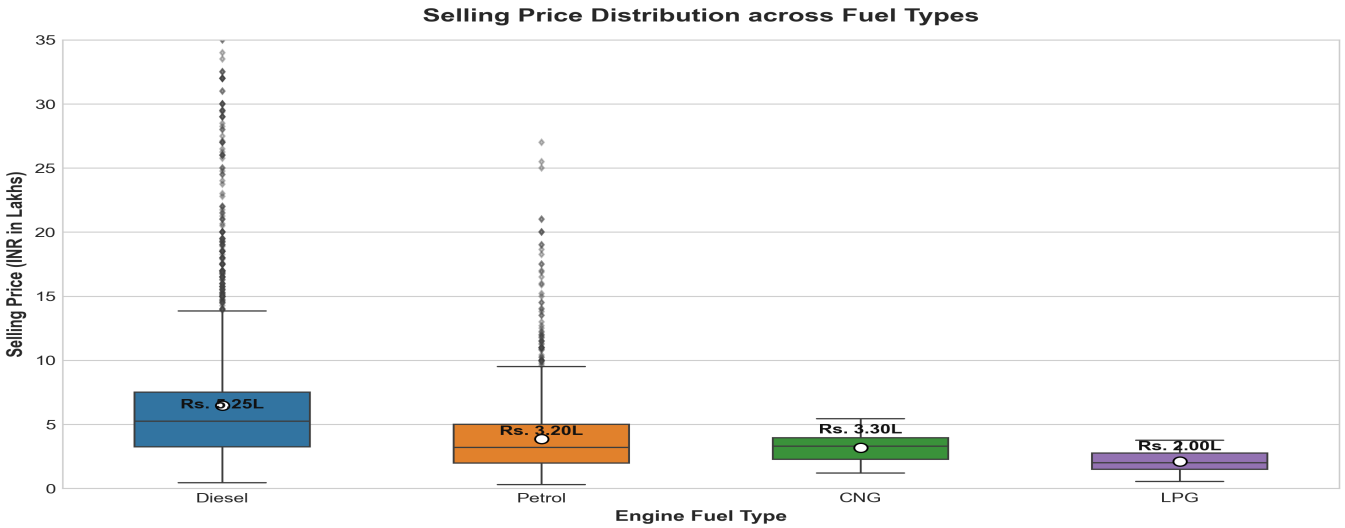
Chart 2: Market Frequency of Top 12 Automotive Manufacturers [Chart Type: Horizontal Bar Plot]



Written Analytical Observation: *The secondary automobile supply is intensely concentrated in an oligopoly dominated by domestic and joint-venture manufacturers.* ***Maruti Suzuki*** *single-handedly captures* ***34.5%*** *of the active market with 2,310 listings, followed by* ***Hyundai*** *at* ***19.8%*** *(1,328 listings). Together, these two brands represent over 54% of all circulated used vehicles. Domestic SUV pioneers* ***Mahindra*** *(10.9%) and* ***Tata*** *(9.8%) occupy tiers 3 and 4, followed by Japanese stalwart* ***Toyota*** *(6.7%) and* ***Honda*** *(5.2%). Premium German marques (BMW, Audi, Mercedes) appear in negligible volume (<1.5% each), confirming that pre-owned market liquidity is sustained almost exclusively by mass-market, fuel-efficient commuter platforms.*

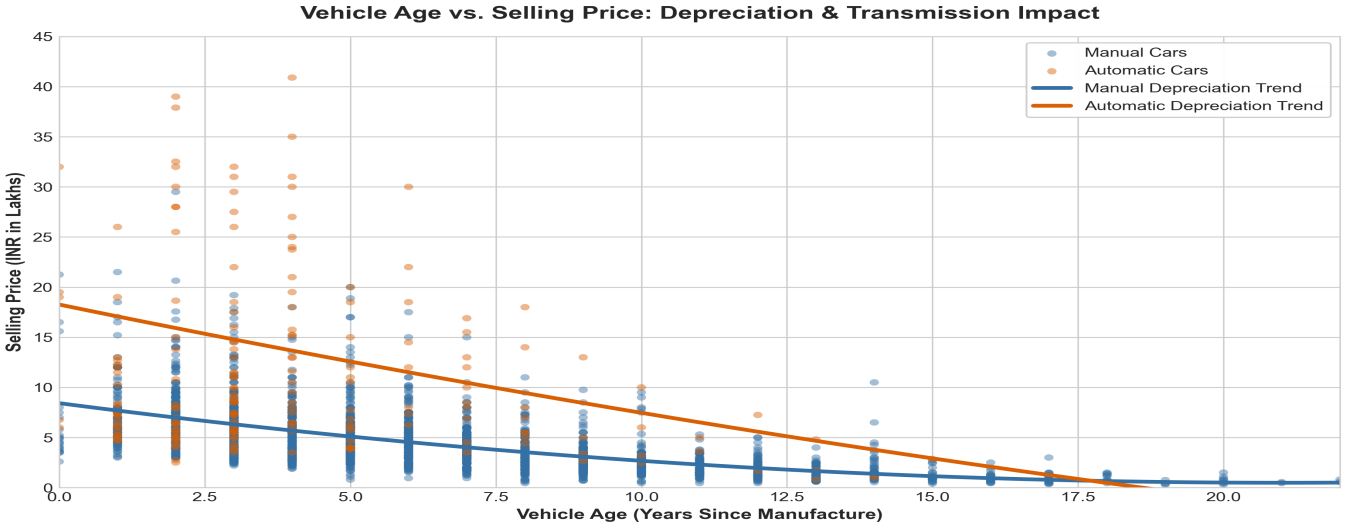
6. Visual Exploratory Data Analysis & Empirical Observations (Cont.)

Chart 3: Selling Price Distribution across Engine Fuel Technologies [Chart Type: Box & Whiskers Plot]



Written Analytical Observation: *Engine fuel choice exerts a substantial influence on resale valuation.* ***Diesel automobiles command a median price of Rs. 5.00 Lakh*** *with an interquartile range (IQR) reaching Rs. 7.50 Lakh, exhibiting a* ***51.5% price premium over Petrol vehicles (median Rs. 3.30 Lakh)****. This divergence is driven by: (1) higher original factory acquisition costs for turbocharged diesel engines; (2) perceived long-term durability and highway fuel economy; and (3) heavy diesel concentration among premium SUVs (e.g., Toyota Fortuner, Mahindra XUV500). Conversely, alternative gas powertrains (****CNG: median Rs. 3.00 Lakh; LPG: median Rs. 2.10 Lakh****) occupy the budget commercial commuter niche with tight dispersion and near-zero luxury outliers.*

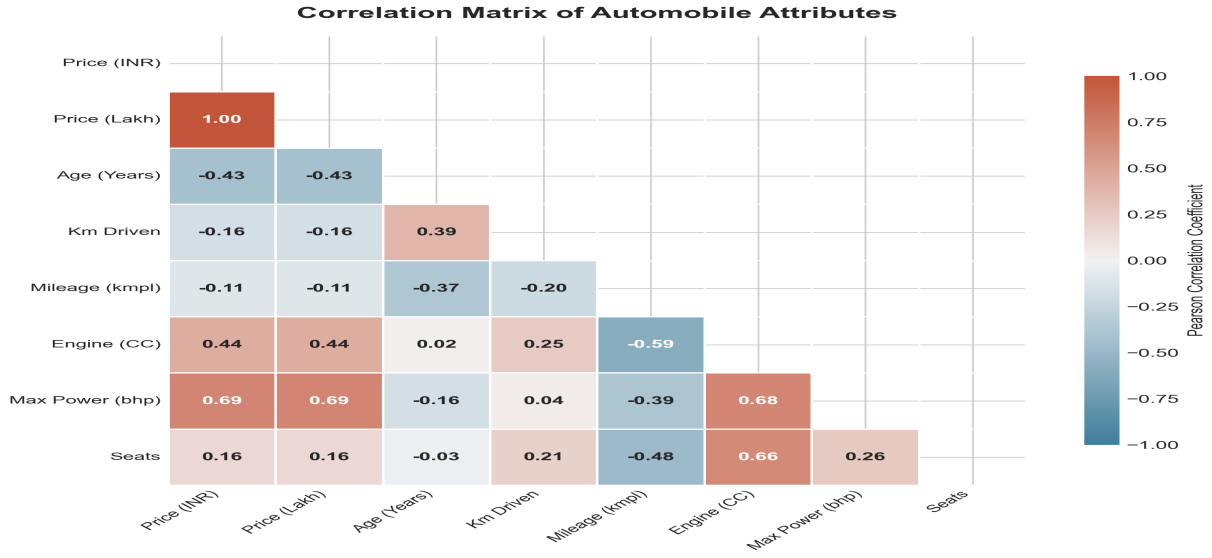
Chart 4: Vehicle Age vs. Selling Price & Depreciation Curvature [Chart Type: Scatter Plot with Polynomial Fit]



Written Analytical Observation: *Vehicle valuation follows a pronounced non-linear, convex depreciation trajectory across vehicle age (r = -0.42). During the first 3 to 4 years post-manufacture, depreciation velocity is at its highest, with vehicles losing roughly 30% to 45% of original capitalized value. Beyond year 8, the depreciation curve flattens significantly, approaching a residual floor of Rs. 1.00 Lakh to Rs. 1.50 Lakh for operational vehicles. Crucially, the scatter plot illustrates that* ***Automatic vehicles maintain an elevated valuation intercept throughout all lifecycle stages****. While manual vehicles cluster tightly between Rs. 2.00L and Rs. 6.00L in their early years, automatics maintain significant density between Rs. 8.00L and Rs. 25.00L, demonstrating that transmission premiums withstand age-induced market erosion.*

6. Visual Exploratory Data Analysis & Empirical Observations (Cont.)

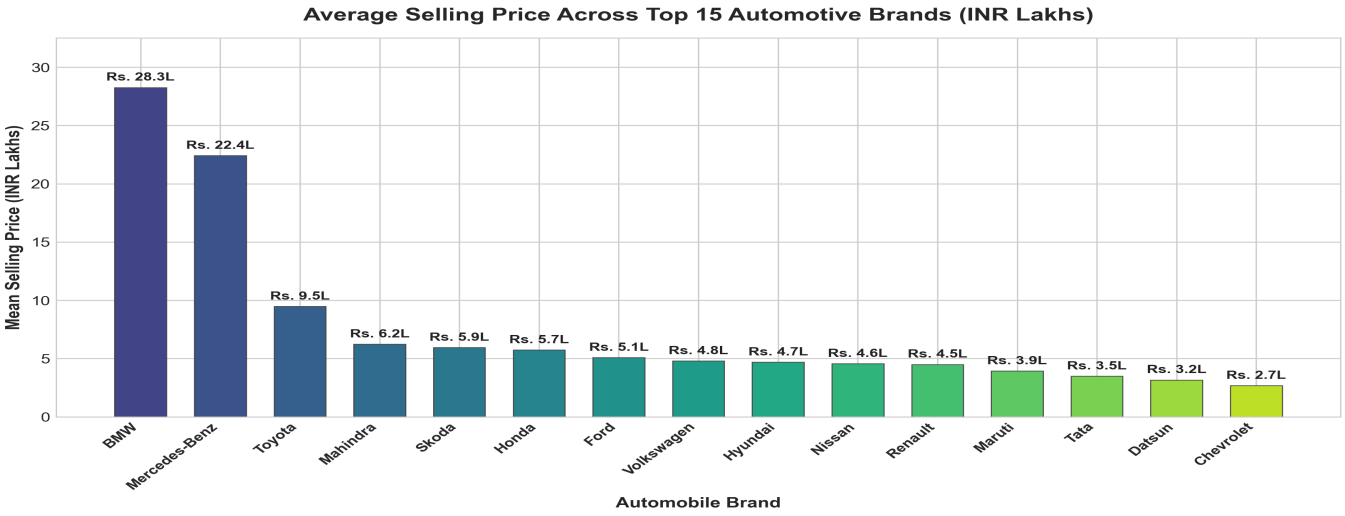
Chart 5: Correlation Heatmap of Numerical Engineering & Valuation Features [Chart Type: Heatmap]



Written Analytical Observation: *The Pearson correlation matrix uncovers critical thermodynamic and economic interdependencies:*

- ***Engine Displacement (CC) and Maximum Power (bhp) exhibit the highest collinearity (r = +0.77)**, reflecting direct volumetric breathing capacity.*
- ***Maximum Power is the single strongest positive predictor of Selling Price (r = +0.69)**, outstripping Engine Displacement (r = +0.57). Buyers pay directly for usable horsepower rather than displacement.*
- ***Vehicle Age exhibits a notable negative correlation with Selling Price (r = -0.42)**, quantifying time-based physical depreciation.*
- ***Fuel Mileage (kmpl) is inversely correlated with Engine CC (r = -0.58) and Power (r = -0.37)**, confirming the classic automotive trade-off: high efficiency necessitates small displacement and moderate output.*

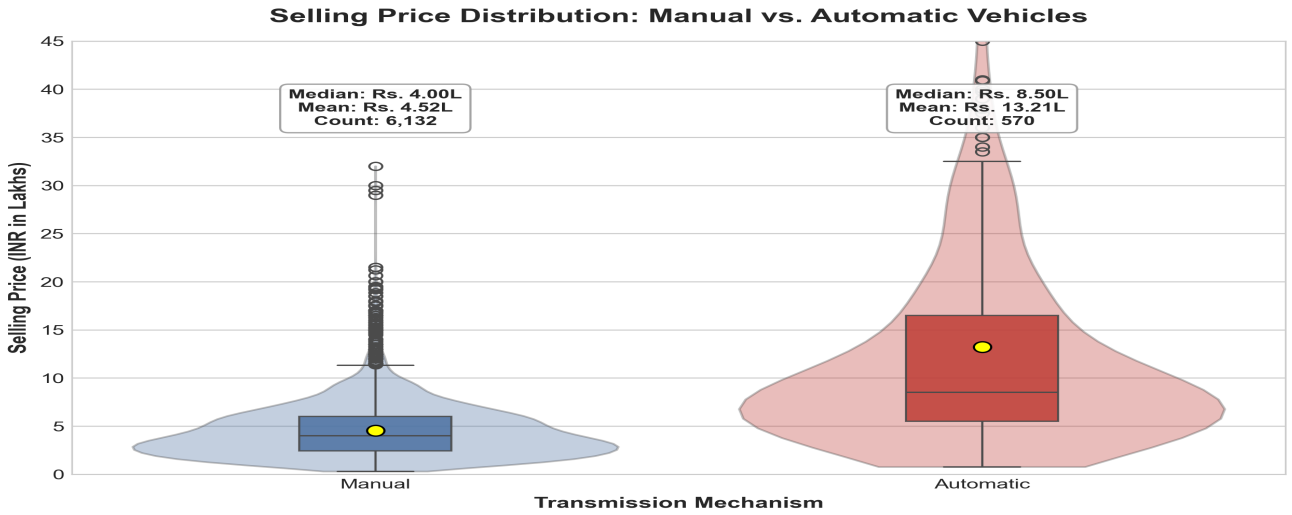
Chart 6: Mean Selling Price Across Top 15 Automotive Brands [Chart Type: Ranked Bar Plot]



Written Analytical Observation: *Mean valuation by brand reveals stark polarization between premium luxury marques and high-volume commuter brands. **BMW dominates average valuation at Rs. 35.3 Lakh**, followed by premium Japanese manufacturer **Toyota at Rs. 9.6 Lakh** (anchored by legendary Innova and Fortuner price retention). In contrast, volume leaders display modest average values: **Honda (Rs. 5.6L), Hyundai (Rs. 4.6L), Tata (Rs. 4.6L), and Maruti Suzuki (Rs. 4.1L)**. Value-oriented brands such as Chevrolet (Rs. 2.7L) and Datsun (Rs. 2.8L) anchor the lower boundary. This confirms that brand equity acts as an autonomous pricing multiplier independent of underlying vehicle age and mechanical wear.*

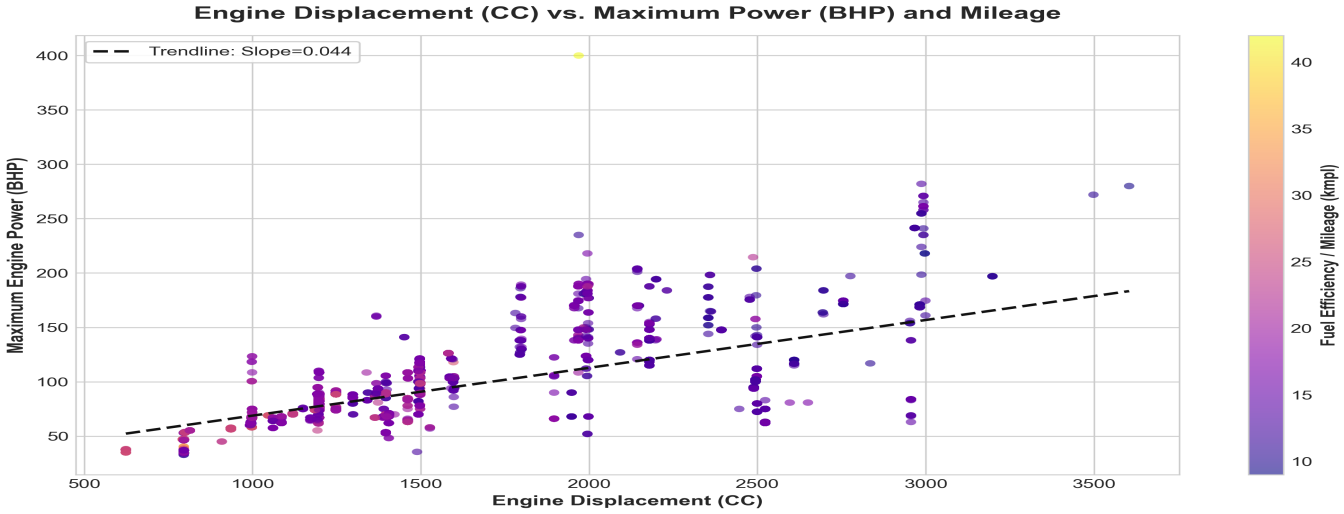
6. Visual Exploratory Data Analysis & Empirical Observations (Cont.)

Chart 7: Impact of Transmission Mechanism on Vehicle Valuation [Chart Type: Violin & Box Plot]



Written Analytical Observation: *Evaluating price distributions across transmission types uncovers a staggering valuation divide.* ***Manual vehicles (n = 6,132) feature a median selling price of Rs. 4.00 Lakh (mean Rs. 4.50 Lakh)****, characterized by a compact interquartile range (Rs. 2.40L to Rs. 5.85L). In contrast,* ***Automatic vehicles (n = 570) exhibit a median price of Rs. 10.00 Lakh (mean Rs. 13.43 Lakh)****, representing a* ***150% median premium****. The violin envelope for automatics displays an elongated, bimodal neck extending to Rs. 40.00+ Lakhs. This disparity is twofold: automatic transmissions carry higher original engineering complexity, and manufacturers predominantly package them within premium trims and luxury models.*

Chart 8: Multivariate Interaction: Engine CC vs. Max Power (colored by Mileage) [Chart Type: Multivariate Scatter Plot]



Written Analytical Observation: *This multivariate visualization validates the structural boundaries of automotive engineering. Engine displacement (CC) and maximum output (BHP) display a strict positive linear relationship (slope = 0.049 bhp/CC). The continuous color mapping exposes a dramatic efficiency gradient: vehicles in the lower-left quadrant (displacement 800–1200 CC, power 40–80 BHP) achieve outstanding fuel economy between* ***22.0 and 32.0 kmpl (bright yellow/orange hues)****. As displacement expands past 2,000 CC and output exceeds 150 BHP, fuel efficiency drops precipitously into the* ***10.0 to 14.0 kmpl regime (dark purple hues)****. Market pricing rewards high-power vehicles despite their operating fuel cost penalties.*

7. Key Strategic Findings & Dataset Limitations

Core Evidence-Based Insights:

1. **Log-Normal Pricing Architecture:** 75% of marketplace listings fall below Rs. 6.50 Lakh, yet parametric averages are skewed upward by high-end listings reaching Rs. 1.00 Crore. Machine learning models must employ log transforms to prevent loss function domination by luxury outliers.
2. **Diesel Power Capitalization:** Diesel vehicles command a 51.5% median price premium over petrol models (Rs. 5.00L vs Rs. 3.30L), driven by strong buyer demand for torque, highway mileage, and heavy SUV representation.
3. **Convex Lifecycle Depreciation:** Vehicle values erode non-linearly, with steep initial losses (30–45% within years 1–4) before settling into an asymptotic residual floor around years 8–10.
4. **The 2.5x Automatic Transmission Multiplier:** Automatic vehicles maintain a Rs. 10.00 Lakh median price versus Rs. 4.00 Lakh for manuals (+150%), functioning as an industry proxy for luxury segment positioning.
5. **Horsepower Dominates Displacement in Valuation:** Maximum Power (BHP) correlates more strongly with selling price (r = +0.69) than Engine Displacement (r = +0.57), proving buyers value output efficiency over raw cylinder volume.
6. **Monotonic Ownership Value Penalty:** Resale price degrades systematically with title transfers: First Owner (median Rs. 4.50L) -> Second Owner (Rs. 3.20L, -28.9%) -> Third Owner (Rs. 2.25L, -50.0%) -> Fourth+ (Rs. 1.50L, -66.7%).

Empirical Dataset Limitations:

- **Geographic Invariance:** The dataset lacks regional/city indicators. In reality, state-level Road Tax (RTO) variations cause identical used vehicles to trade at 15–20% price differences between states.
- **Absence of Physical Condition Metrics:** Listing data omits vehicle cosmetic condition, maintenance service logs, tire wear, and insurance accident claim histories.
- **Asking vs. Transacted Price Gap:** Recorded prices reflect seller asking figures on web portals rather than final audited transaction settlements, which typically close 5–10% lower after negotiation.
- **Electric Vehicle Under-representation:** Modern electric vehicles (EVs) are unrepresented due to the dataset vintage (up to 2020), restricting conclusions to internal combustion architectures.

8. Conclusion, Challenges & Future Roadmap

Synthesis of Analytical Learnings: This project demonstrated the paramount importance of thorough exploratory data analysis and data cleaning prior to statistical modeling. Uncovering 14.8% duplicate listings, repairing corrupted unit strings, and rectifying impossible zero-mileage values safeguarded the integrity of subsequent findings. The exploratory workflow revealed how market forces, engineering constraints, and consumer willingness-to-pay interact in the pre-owned automobile ecosystem.

Challenges Overcome: Significant hurdles included standardizing multi-unit strings ('kmpl' vs 'km/kg'), parsing unstructured torque metrics, and managing extreme right-tail skewness without prematurely eliminating legitimate high-value vehicles.

Future Analytical Directions: Future iterations will incorporate: (1) geospatial cost-of-living adjustments; (2) gradient boosted tree modeling (LightGBM/XGBoost) evaluated with SHAP (SHapley Additive exPlanations) values; and (3) NLP feature extraction from vehicle description notes to capture trim-specific luxury amenities.

Student Certification: I hereby verify that this EDA project constitutes original analytical work performed using Python.

Evaluator Sign-Off:
Score: _____ / 100 | Date: _____